Trinity College Dublin
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The University of Dublin

Genomic & Epidemiological Surveillance of Bacterial Pathogens

22-26 June, 2026

What Does a Bacterial Genome Look like?

Caterina Guzmán-Verri, PhD

PIET-UNA SALUD, Veterinary School, Universidad Nacional, Costa Rica

catguz@una.cr

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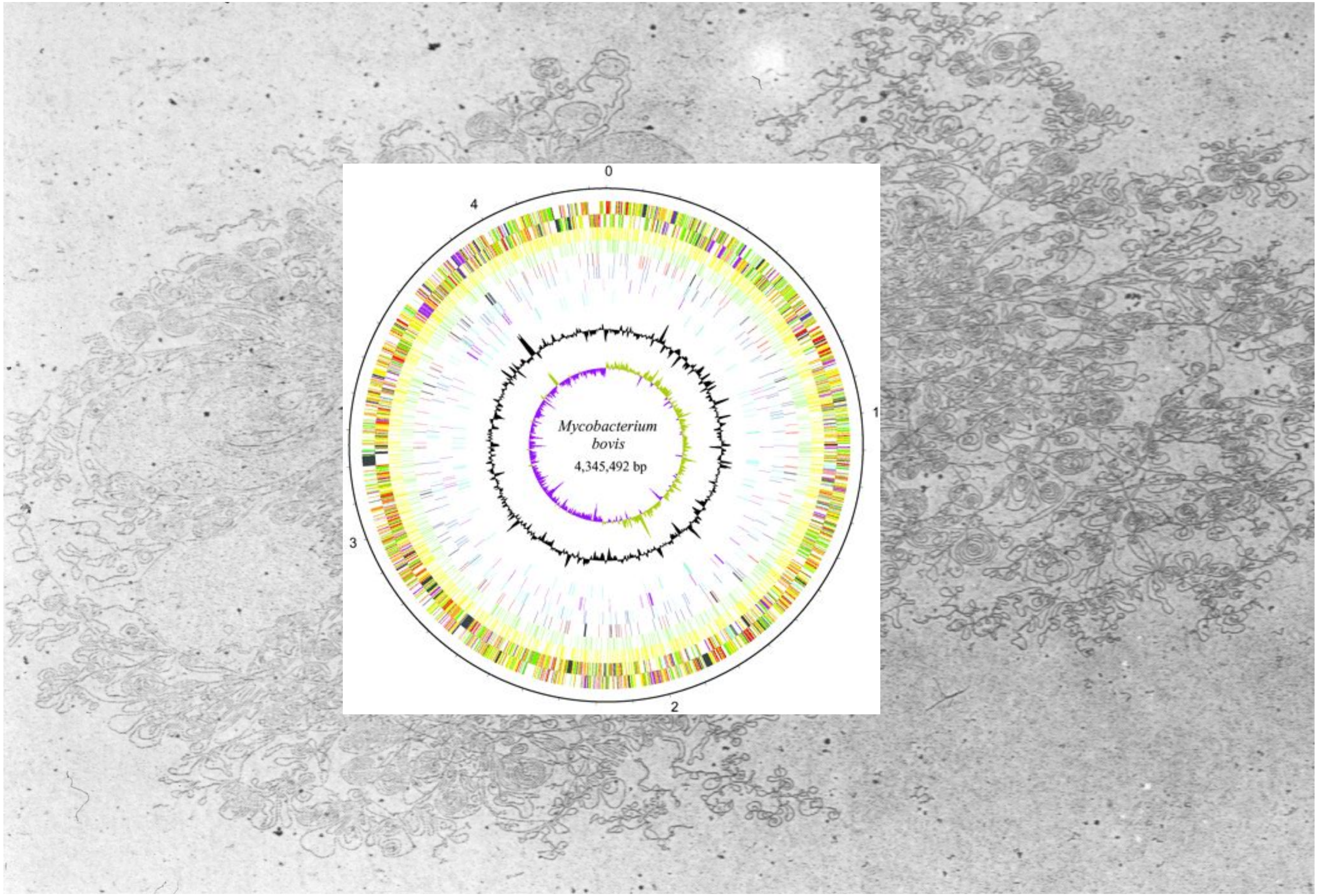
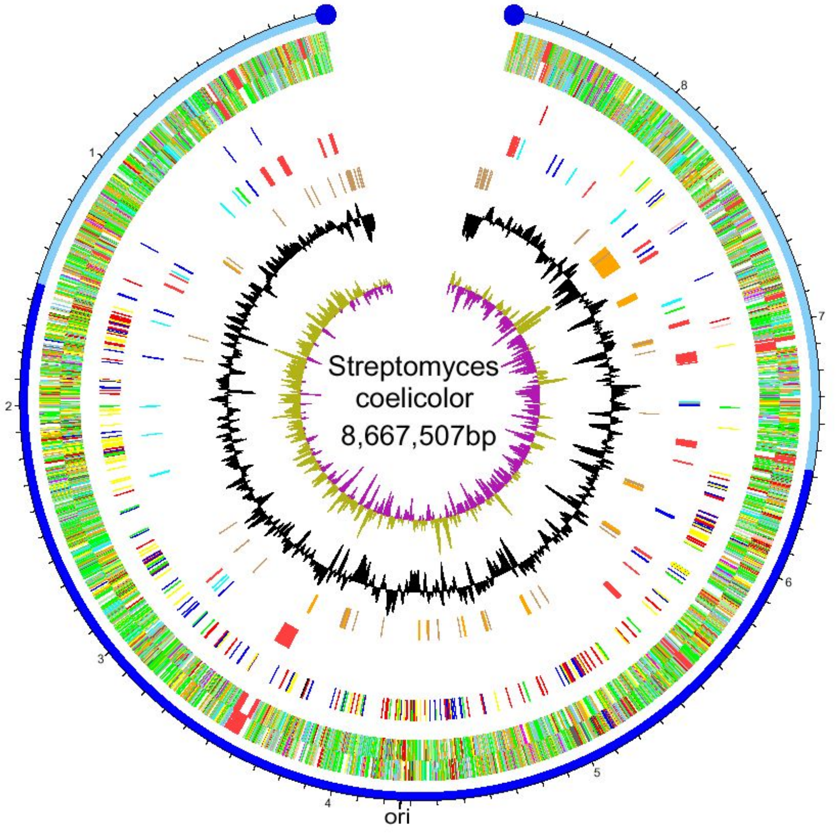
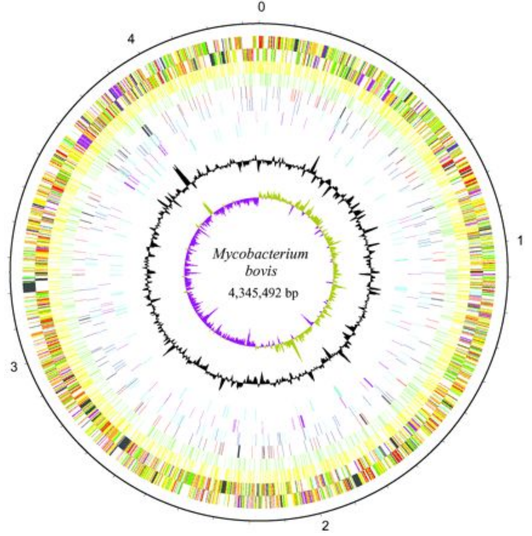
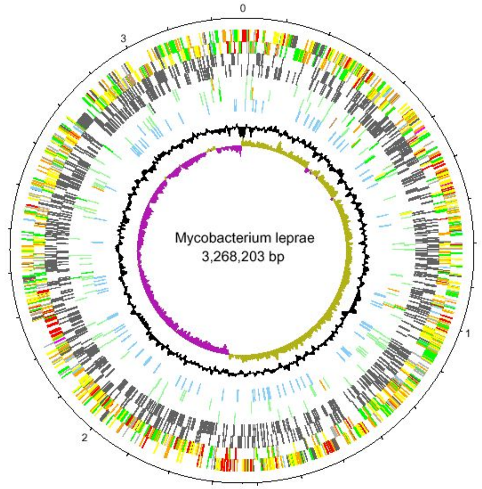
Interpreting the DNA

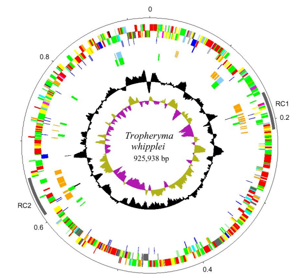


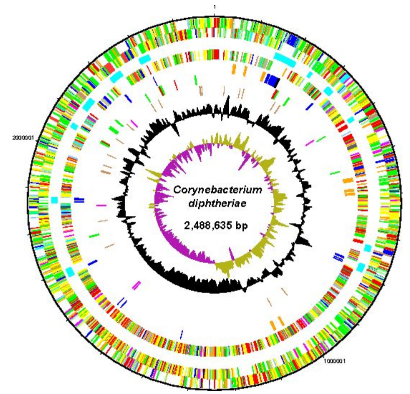




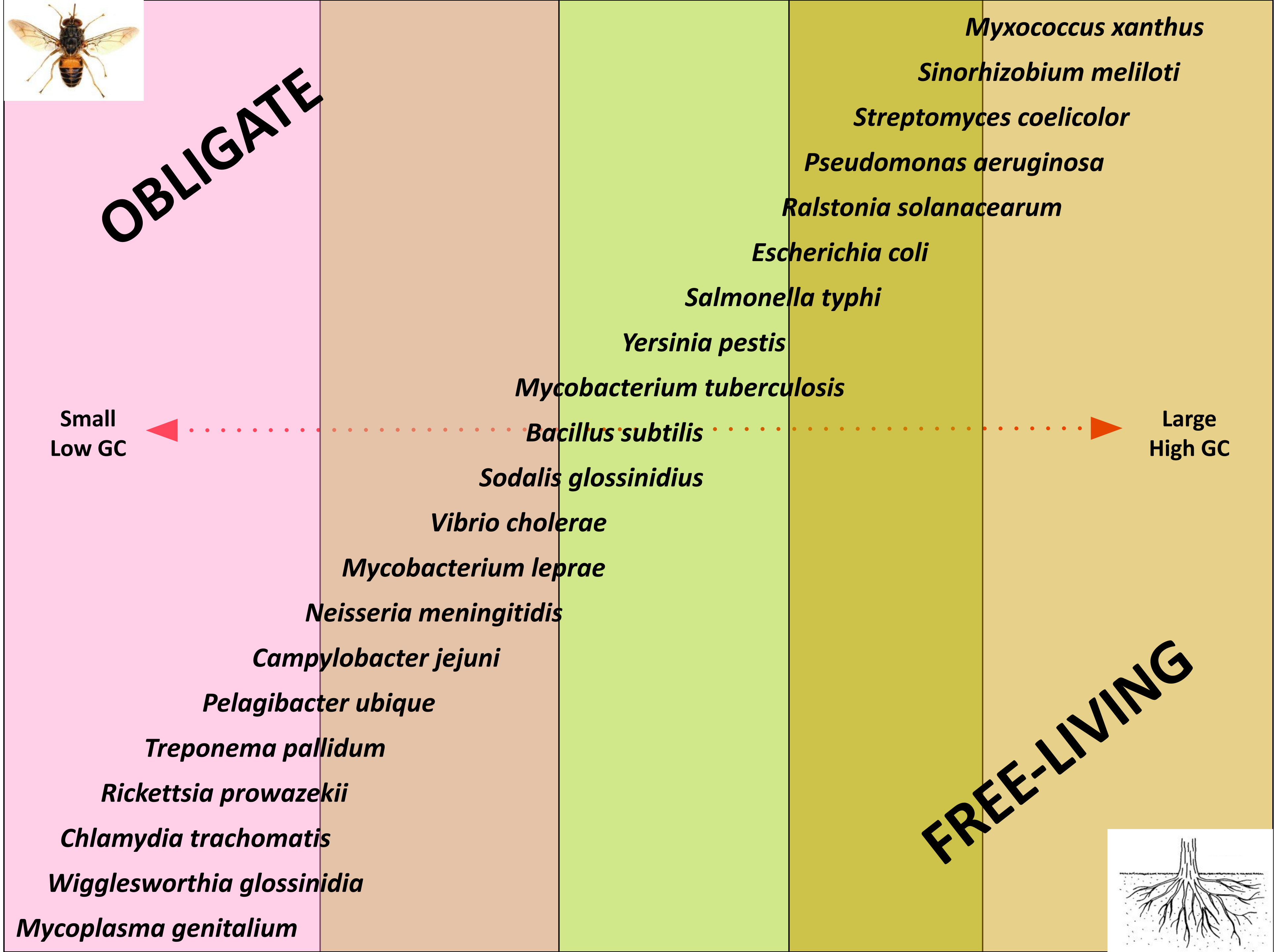


Linear/circular
Range of sizes
Huge variation

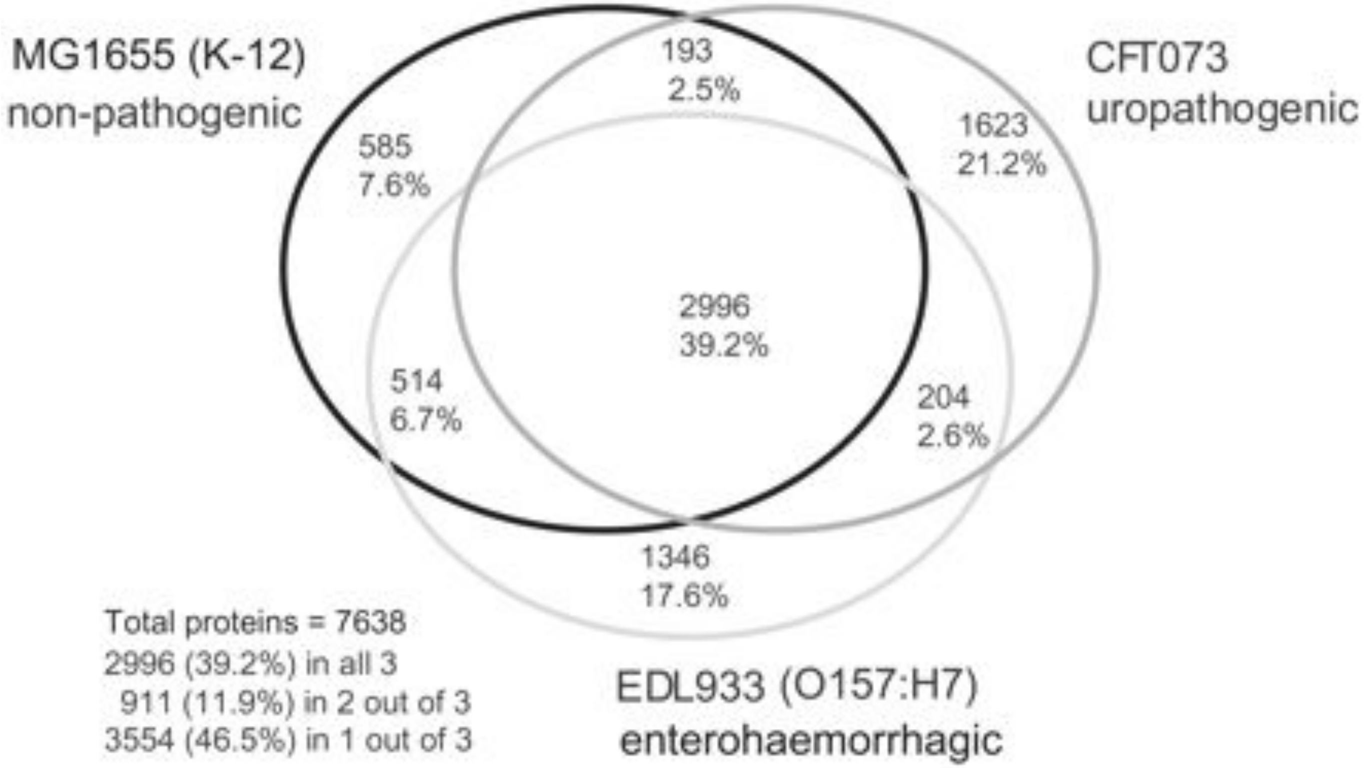




Eukaryotes	*Schistosoma*	(Helminth worm)	270 Mb
	Malaria	(Protozoa)	25 Mb
	Candida	(Fungi)	16 Mb
Prokaryotes	*Streptomyces*		9 Mb
	Yersinia		5 Mb
	MRSA		3 Mb
	Chlamydia		1 Mb
Plasmids	R478		290 Kb
Bacterial Viruses	Vi		40 Kb


OBLIGATE
Myxococcus xanthus
Sinorhizobium meliloti
Streptomyces coelicolor
Pseudomonas aeruginosa
Ralstonia solanacearum
Escherichia coli
Salmonella typhi
Yersinia pestis
Mycobacterium tuberculosis
Small
Low GC
Bacillus subtilis
Large
High GC
Sodalis glossinidius
Vibrio cholerae
Mycobacterium leprae
Neisseria meningitidis
Campylobacter jejuni
Pelagibacter ubique
Treponema pallidum
Rickettsia prowazekii
Chlamydia trachomatis
Wigglesworthia glossinidia
Mycoplasma genitalium
FREE-LIVING

Protein coding genes shared between three of the sequenced *E. coli*.



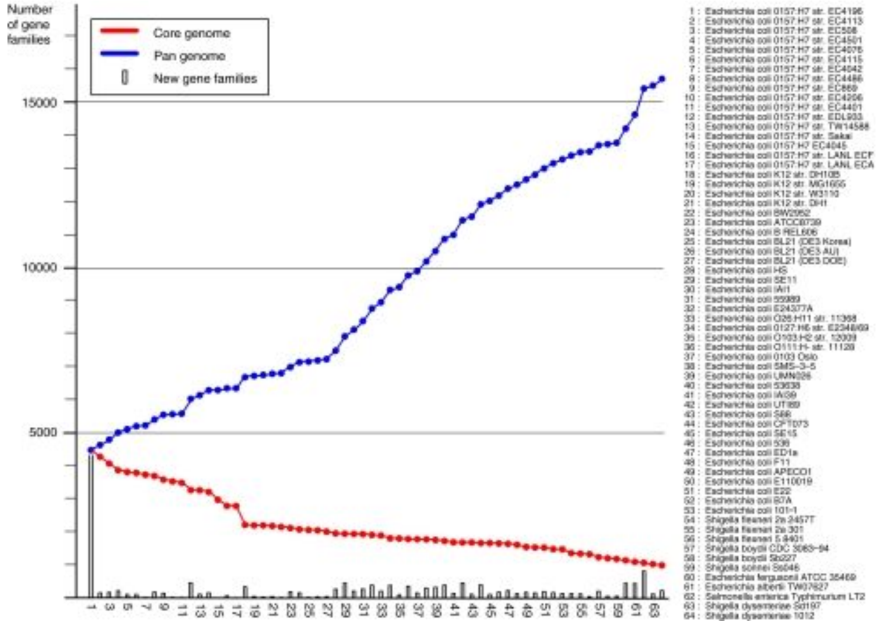
Pangenome:
All genes, constant or variable, that are present in a species.

Core genome:
Genes present in all or the majority of the members of a species and close relatives.
They are usually considered to be essential for growth and replication.

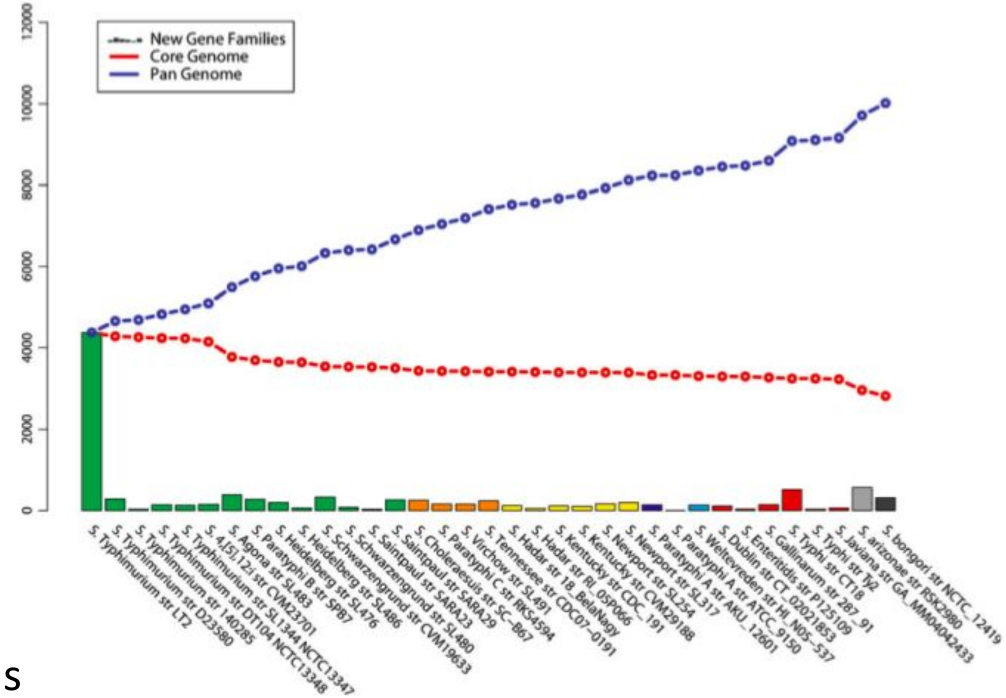
Accessory genome:
Set of genes that are variably present in members of a specie. May confer an advantage to a particular isolate in a specific niche.

E. coli and *Salmonella* Pangenomes

E. coli



Salmonella

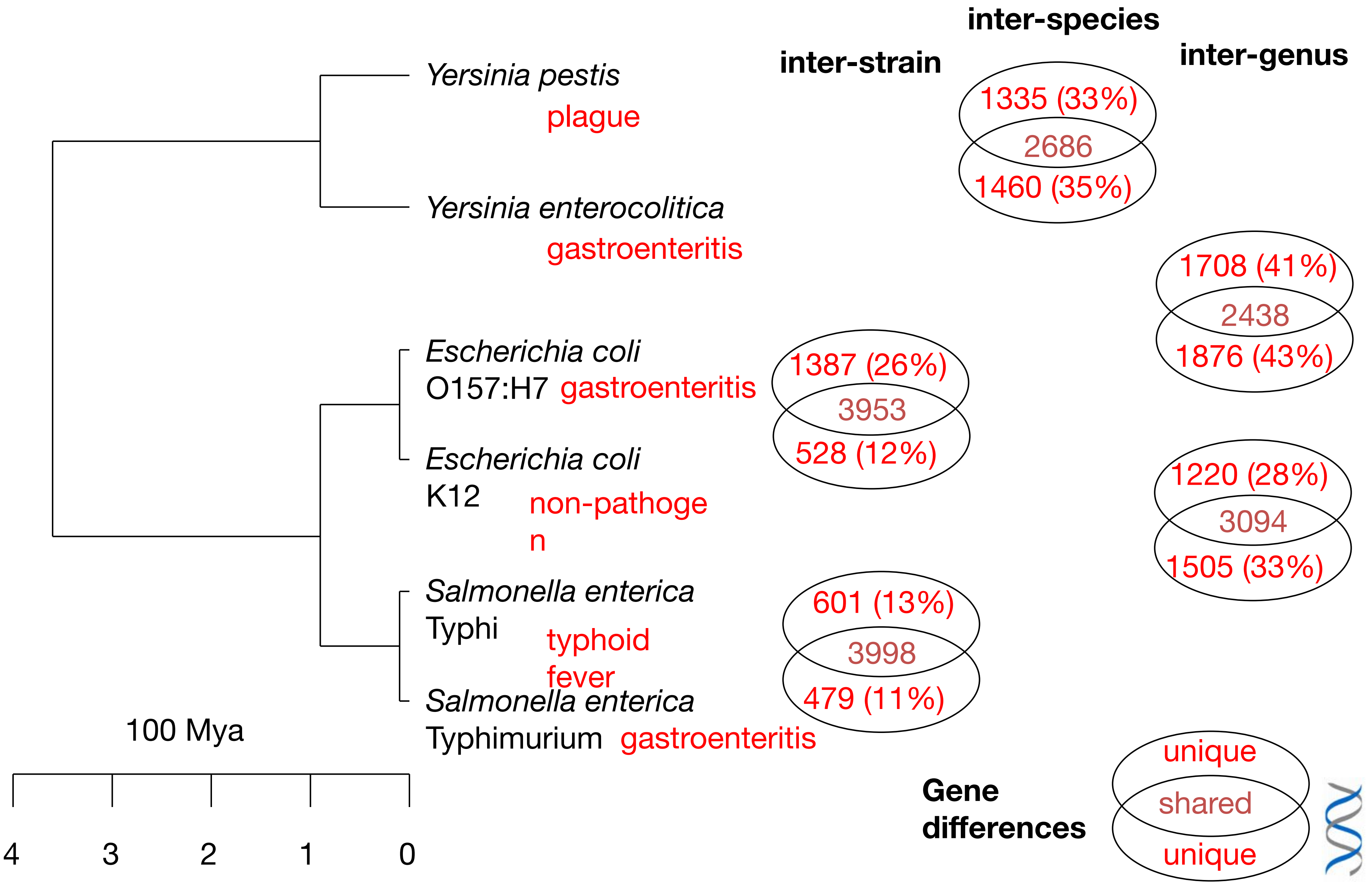


- 17 genomes genome size 5020 +/- 446 cds
- core ~2200 CDS's
- 300 novel genes/ additional genome
- open pangenome still evolving by acquisition
- gene reservoir >13k cds's

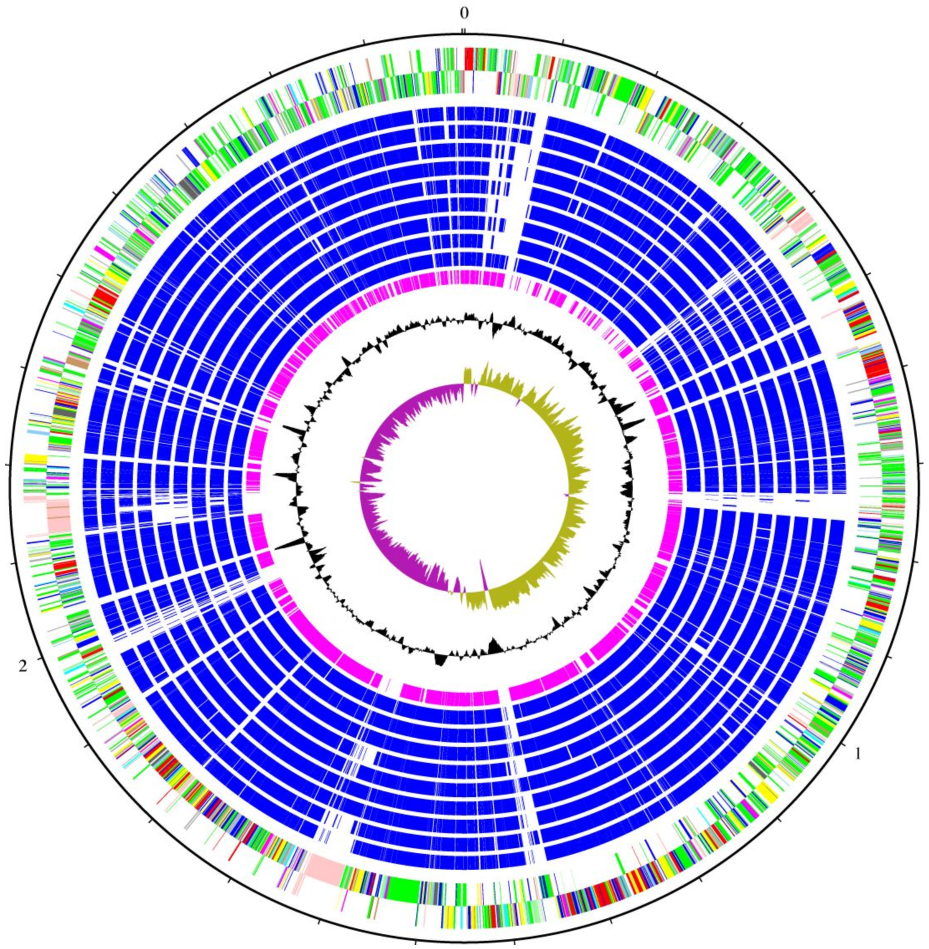
35 genomes core 2811
accessory >10,000

Rasko et al., 2008; Jacobsen et al., 2011

Bacterial diversity the variable gene pool



How can we use a pathogen's genome?



- Unravel the mechanisms of disease
  - Highlight new virulence factors
  - Reconstruct metabolism
- Identify new targets for antimicrobial therapies
  - Rational drug design
  - Identification of potential vaccine targets

WGS to study bacterial disease epidemiology

Genomic variability between strains = meassuring distances by:

- Determining SNPs in core genome aligments,
- And/or by indexing allelic variation in core genes assigning types to unique allelic profiles.

CONTEXT IS VERY IMPORTANT (interpretation of distances is *highly* organism specific)

The inclusion of accessory genes or plasmid sequences might help to get the discrimination power.

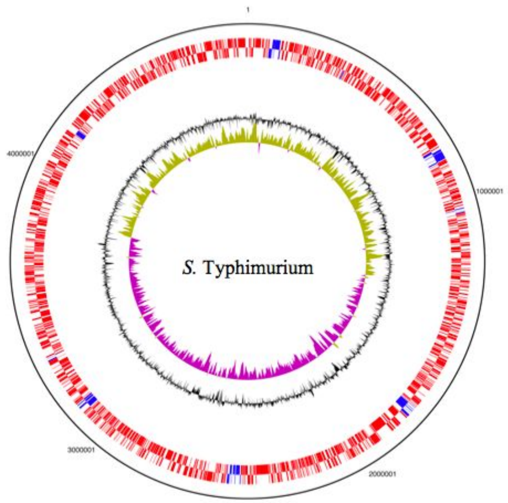
Sources of variability between bacterial genomes

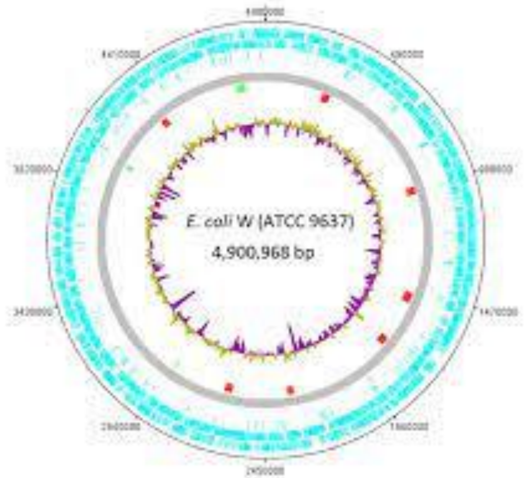
-Point mutations (SNPs, single nucleotide indels. Mutation rates: *M. tuberculosis* 4 SNPs/genome/4 years, *H. pylory*: 30 SNPs/genome/year

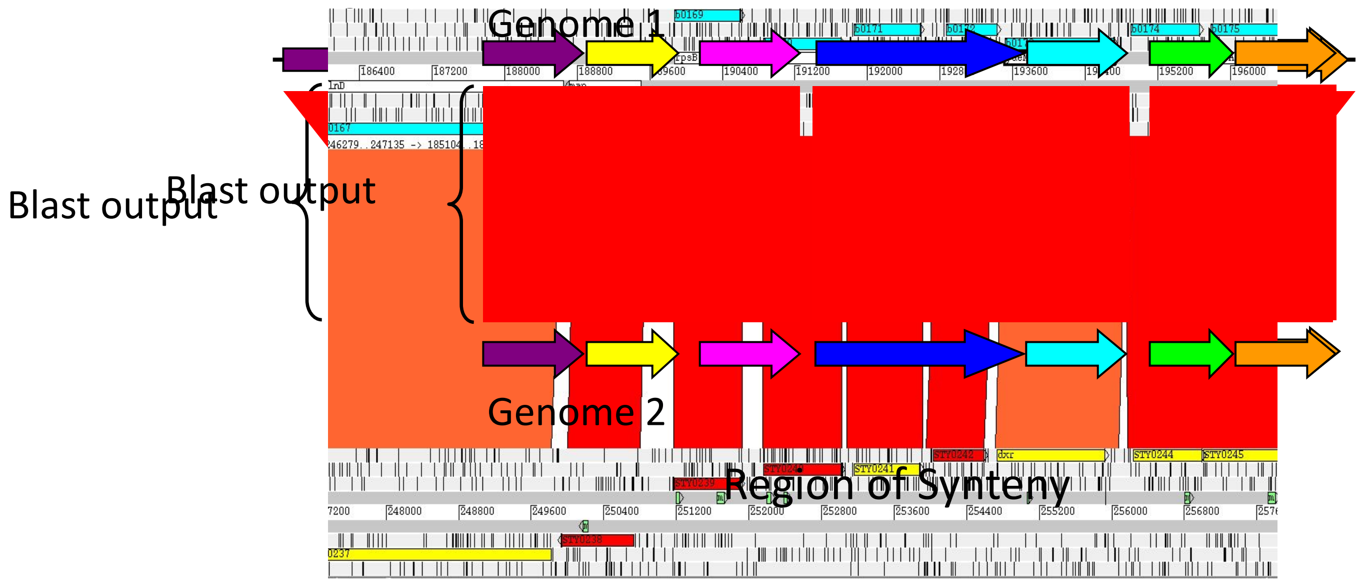
-Homologous recombination

- Differences in genome content (large indels, genome rearrangements and transfer of exogenous DNA and extrachromosomal elements (plasmid, phages)

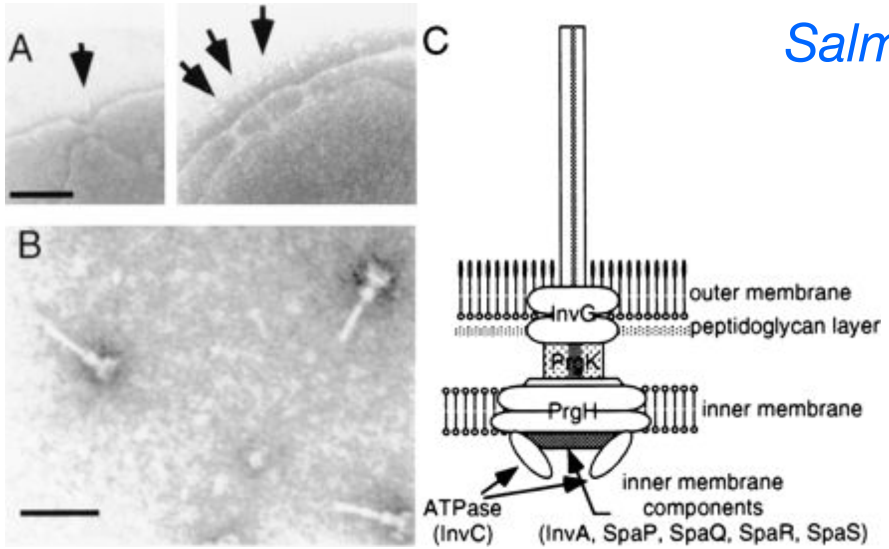
Comparative Genomics



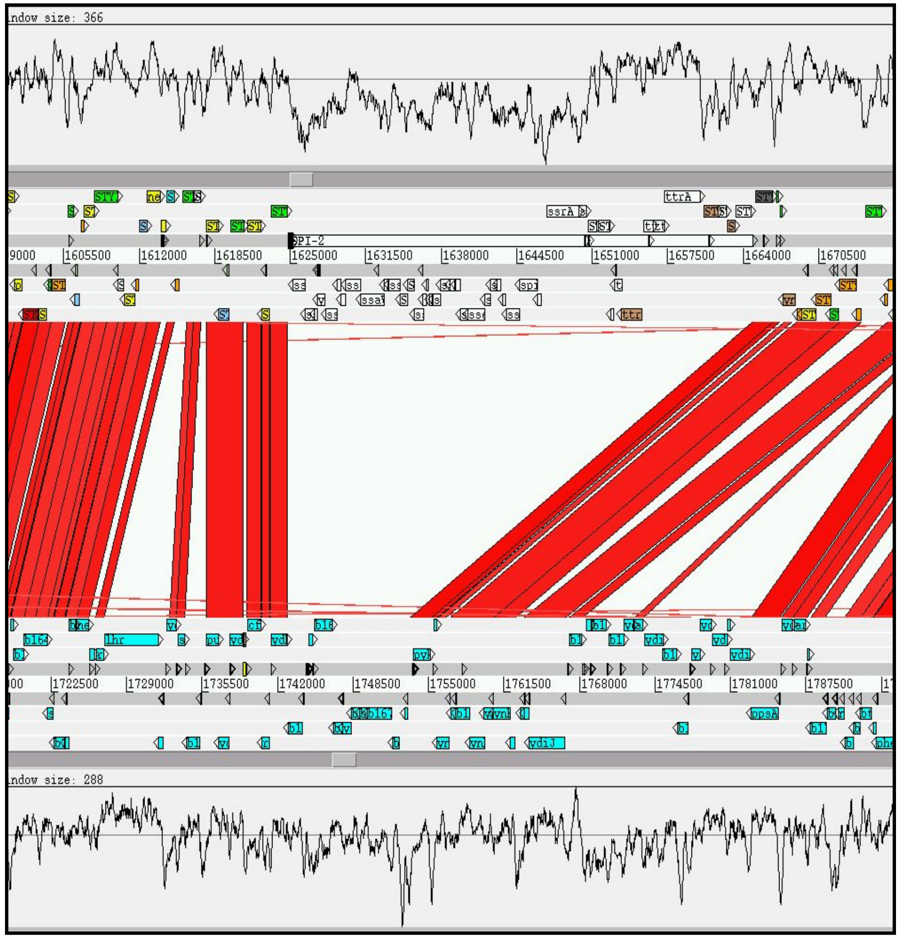




Salmonella pathogenicity Islands SPI



G+C
tRNA
phage/IS genes
Pseudogenes
virulence



Salmonella Pathogenicity Islands

Nomenclature	Description	size & features	Original Reference
SPI-1	T3SS, iron uptake	40 kb	(Hansen-Wester and Hensel 2001)
SPI-2	T3SS	40 kb, tRNA-*valV*	(Hensel 2000)
SPI-3	*mgtCB* Mg^{2+} transport	17 kb, tRNA-*selC*	(Blanc-Potard et al. 1999)
SPI-4	type I secretion and large repetitive protein	23 kb	(Parkhill et al. 2001a; Wong et al. 1998)
SPI-5	T3SS effectors	8 kb, tRNA-*serT*	(Wood et al. 1998)
SPI-6	*saf tcs* fimbrial systems	59 kb, tRNA-*aspV*	(Parkhill et al. 2001a)
SPI-7	Vi antigen and the SopE prophage	134 kb, tRNA-*pheU*	(Parkhill et al. 2001a)
SPI-8	bacteriocin immunity protein	6.8 kb, tRNA-*pheV*	(Parkhill et al. 2001a)
SPI-9	type I secretion and large repetitive protein	16kb	(Parkhill et al. 2001a)
SPI-10	prophage ST46, *sef* fimbrial operon	33 kb, tRNA-*leuX*	(Parkhill et al. 2001a)
SPI-11	T3SS effectors and PhoPQ-activated proteins	14 kb	(Chiu et al. 2005)
SPI-12	*msgA* & *narP*	6.3 kb	(Chiu et al. 2005)
SPI-13	required for survival in chicken macrophages	~7 kb, tRNA-*pheV*	(Shah et al. 2005)
SPI-14	unknown	~11 kb	(Shah et al. 2005)
SPI-15	Proteins of unknown function	6.5kb tRNA-*gly*	(Vernikos and Parkhill 2006)
SPI-16	bacteriophage remnants and LPS modification genes	4.5kb tRNA-*arg*	(Vernikos and Parkhill 2006)
SPI-17	bacteriophage remnants and LPS modification genes	5.1kb tRNA-*arg*	(Vernikos and Parkhill 2006)
SGI-1*	Multiple antibiotic resistance genes	43 kb	(Boyd et al. 2001)
HPI	iron uptake in *Yersinia*	ND	(Oelschlaeger et al. 2003)

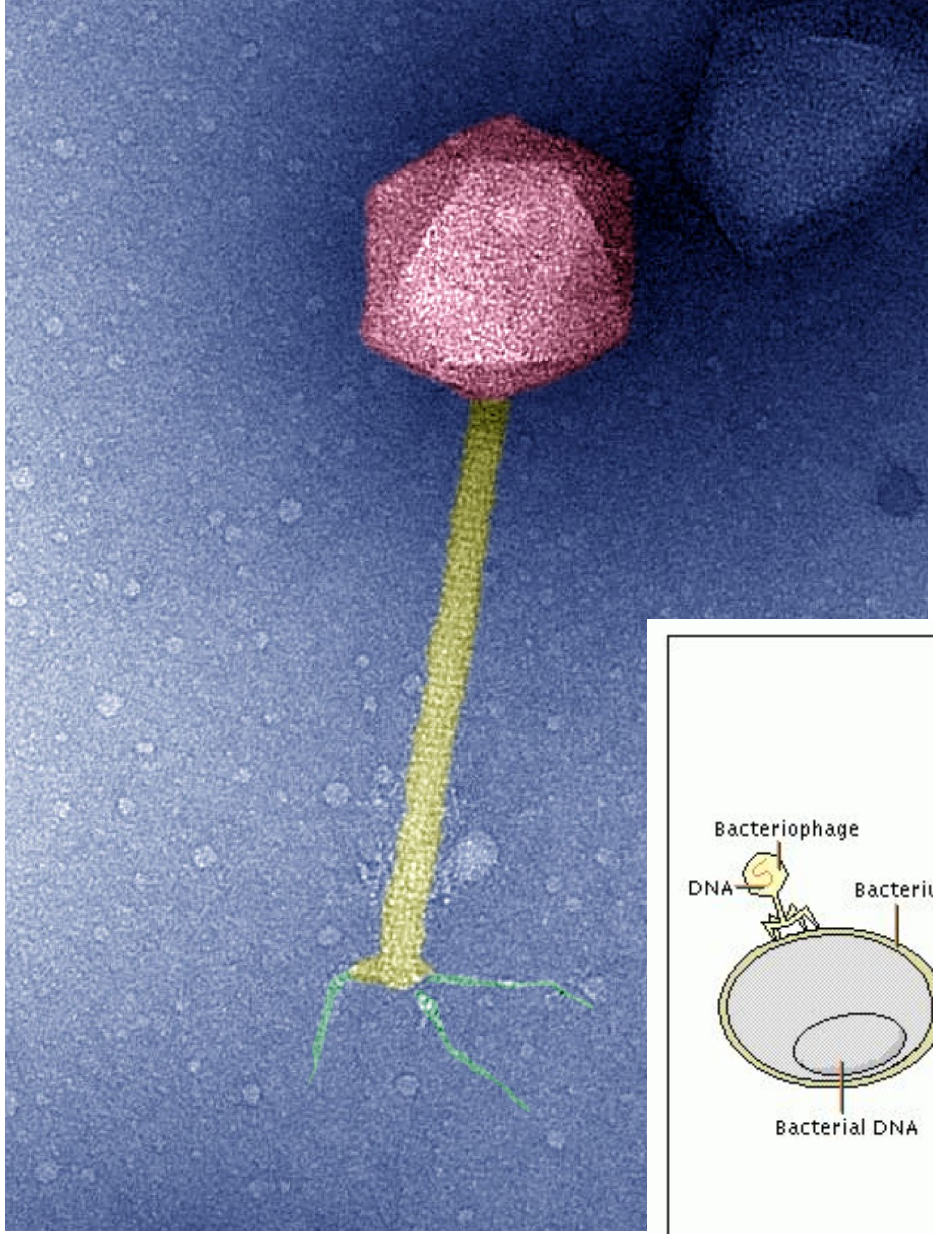
* *S.* Typhimurium DT104 only
ND – not determined

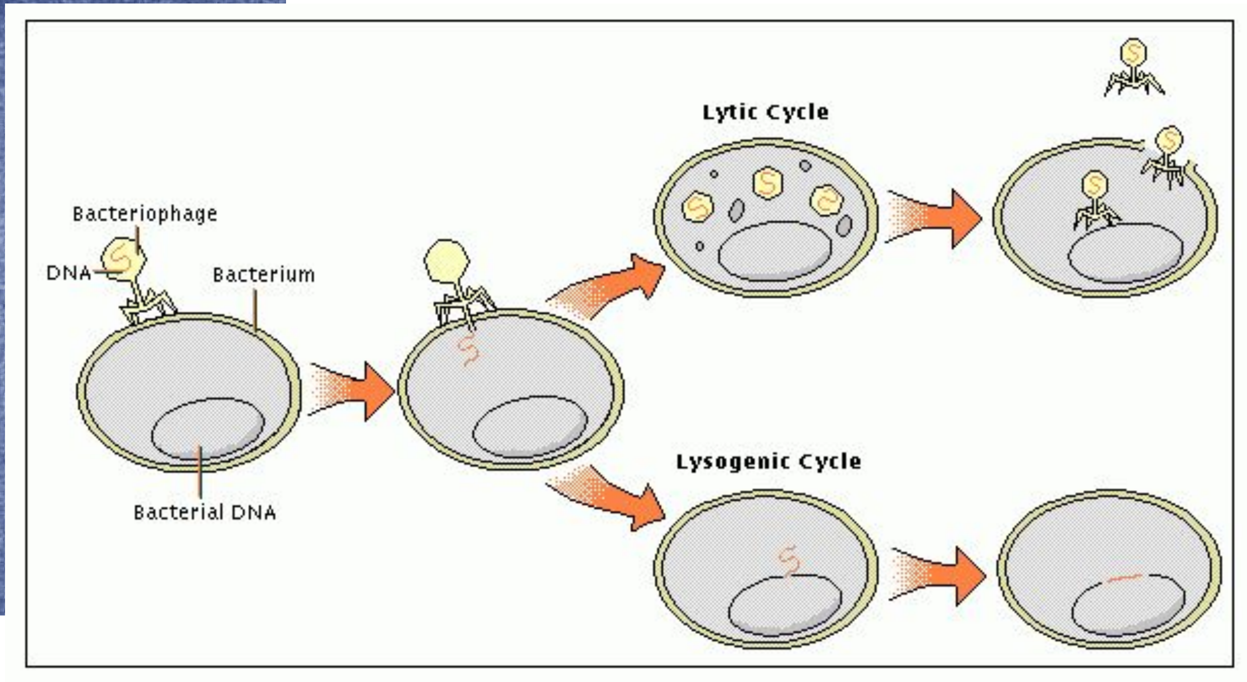
Mechanisms of Horizontal Gene Transfer

Bacteriophage:

- Injects DNA directly into the bacterial host

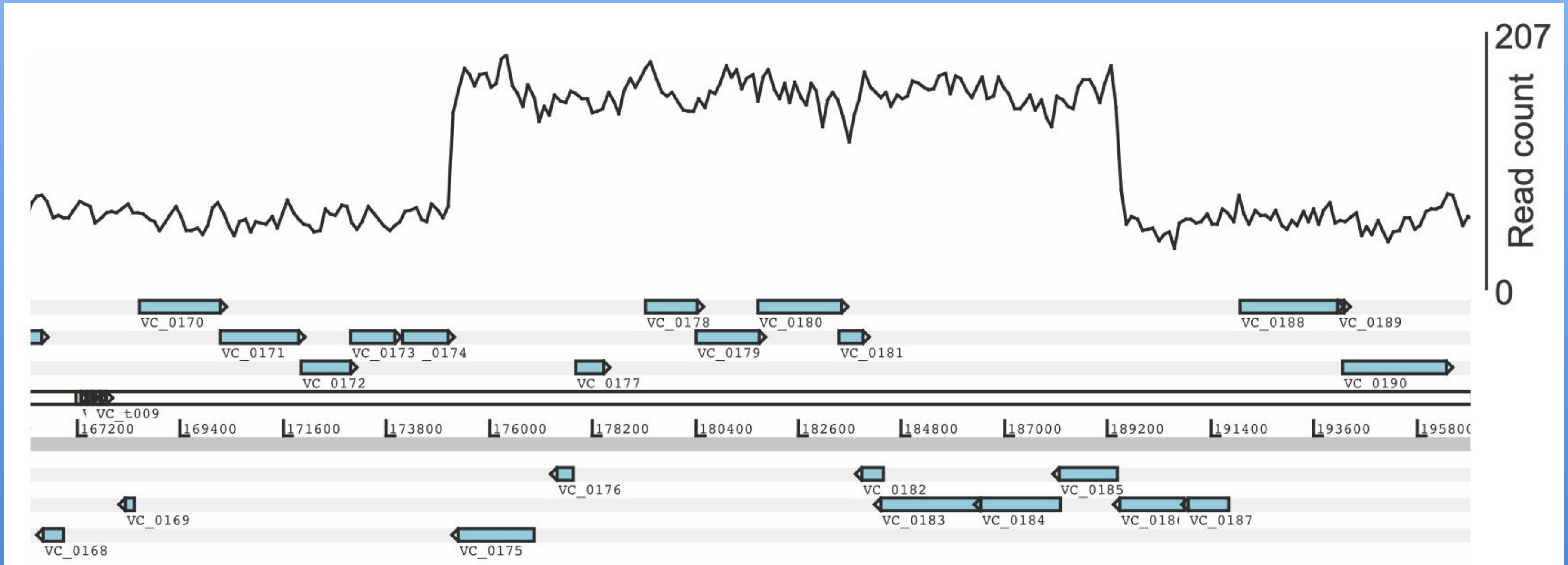
- 2 lifestyle options:
  Lytic
  Lysogenic





Lysogenic conversion

What's wrong with this picture?



PMID 30971707

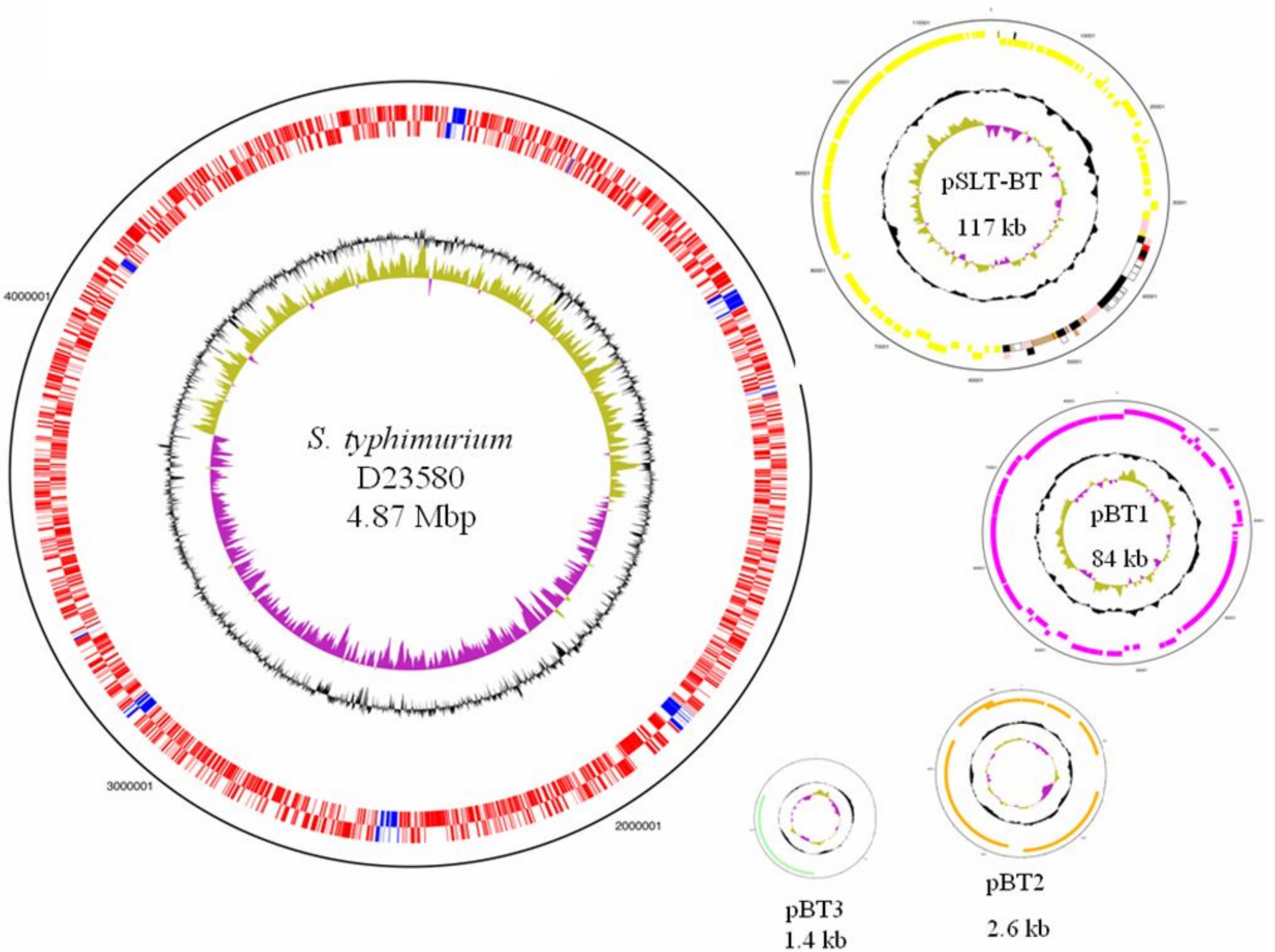
Bashai et al., 1988

Phage carried virulence factors

Protein	Gene	Phage	Organism
Extracellular toxins			
Diphtheria toxin	*tox*	ß-Phage	*C. diphtheriae*
Cholera toxin	*ctxAB*	CTXφ	*V. cholerae*
Proteins altering antigenicity			
Membrane proteins	Mu-like	Pnm1	*N. meningitidis*
O-antigen acetylase	*oac*	Sf6	*S. flexneri*
Effector proteins involved in invasion			
Type III effector	*sopE*	SopEf	*S. enterica*
Type III effector	*sseI* (*gtgB*)	GIFSY-2	*S. enterica*
Enzymes			
Superoxide dismutase	*sodC*-I	GIFSY-2	*S. enterica*
Neuraminidase	*nanH*	Fels-1	*S. enterica*
Serum resistance			
OMPb	*bor*	λ	*E. coli*
OMP	*eib*	λ-like	*E. coli*
Adhesions for bacterial host attachment			
Vir	*vir*	MAV1	*M. arthritidis*
Phage coat proteins	*pblA* , *pblB*	SM1	*S. mitis*
Others			
Mitogenic factor	*toxA*	Unnamed	*P. multocida*
Antivirulence	*grvA*	GIFSY-2	*S. enterica*

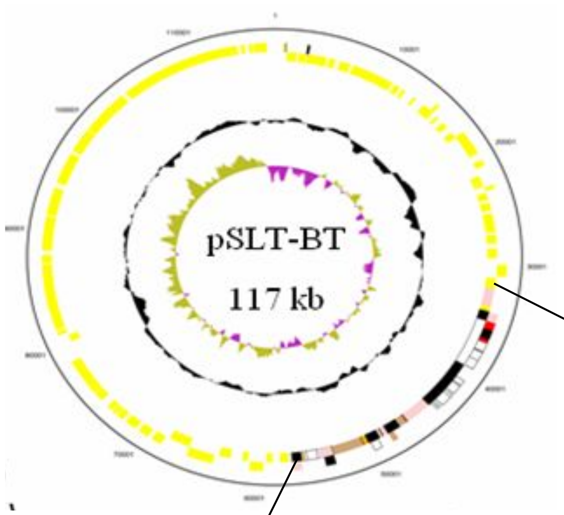
Adapted from Boyd & Brussow (2002) *Trends Microbiol*.

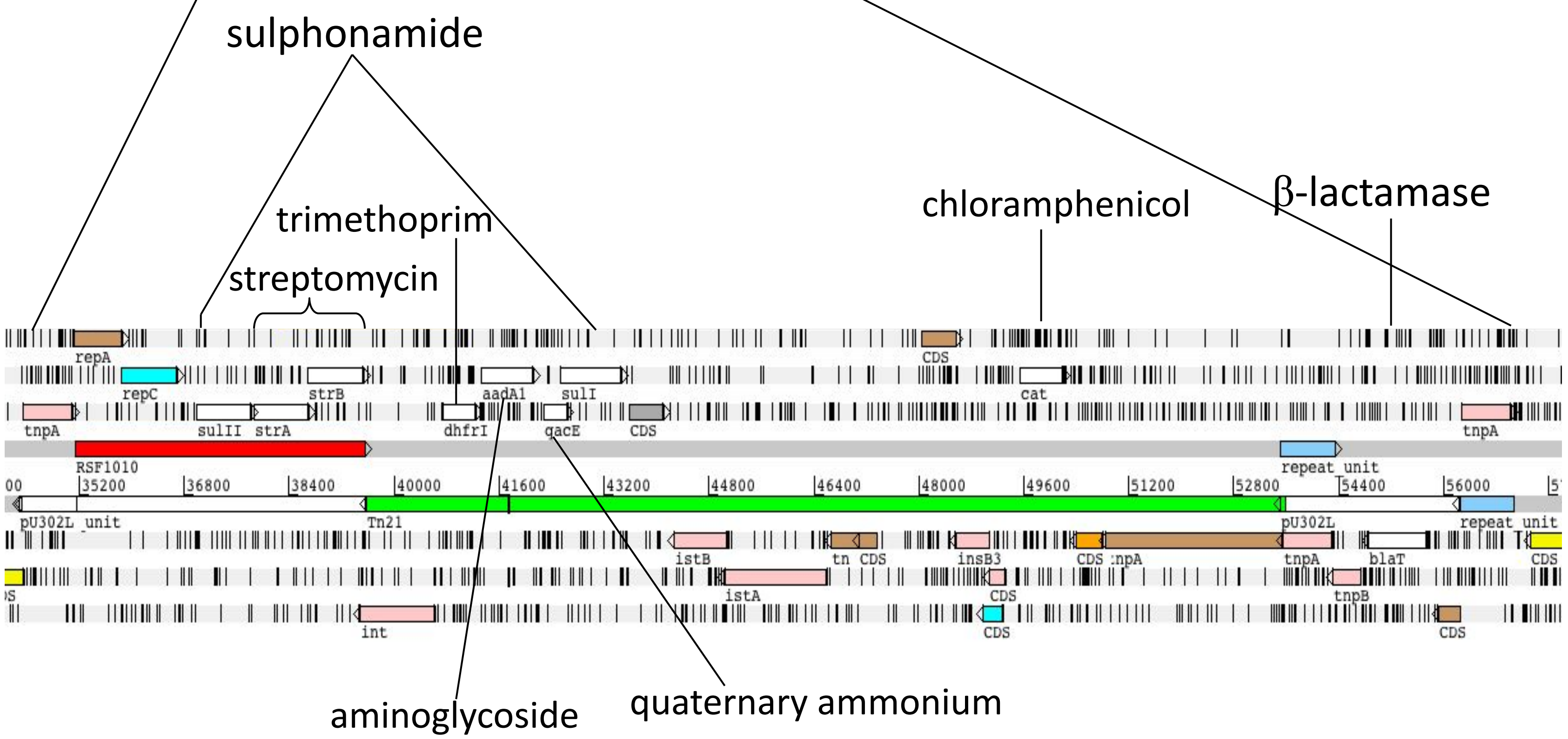
Extra chromosomal accessory elements: Plasmids



S. typhimurium plasmid carrying:

- Genes essential for virulence/invasiveness
- Multiple drug resistance genes





Genome architecture
– replicons and compartments

Species	Appellation	Size (kb)	Shape	rDNA no.
Streptomyces coelicolor	Chromosome	8667	Linear	6
	Plasmid	356	Linear	0
	Plasmid	31	Circular	0
Agrobacterium tumefaciens	Chromosome	2842	Circular	2
	Chromosome	2057	Linear	2
	Plasmid	543	Circular	0
	Plasmid	214	Circular	0
Borellia burgdorferi	Chromosome	911	Linear	1
	Plasmid (n = 11)	9–54	Circular/Linear	0
Brucella melitensis	Chromosome	2117	Circular	2
	Chromosome	1178	Circular	1
Clostridium acetobutylicum	Chromosome	3941	Circular	11
	Megaplasmid	192	Circular	0
Deinococcus radiodurans	Chromosome	2649	Circular	3
	Chromosome	412	Circular	0
	Megaplasmid	177	Circular	0
	Plasmid	46	Circular	0
Ralstonia solanacearum	Chromosome	3716	Circular	3
	Megaplasmid	2095	Circular	1
Salmonella typhi	Chromosome	4809	Circular	7
	Plasmid	218	Circular	0
	Plasmid	107	Circular	0
Sinorhizobium meliloti	Chromosome	3654	Circular	3
	Megaplasmid	1683	Circular	0
	Megaplasmid	1354	Circular	0
Vibrio cholerae	Chromosome	2941	Circular	8
	Chromosome	1072	Circular	0
Yersinia pestis	Chromosome	4654	Circular	6
	Plasmid (n = 3)	10–96	Circular	0

From Ochman (45).

nature > nature communications > articles > article

Article | Open access | Published: 11 August 2025

Giant extrachromosomal element "Inocle" potentially expands the adaptive capacity of the human oral microbiome

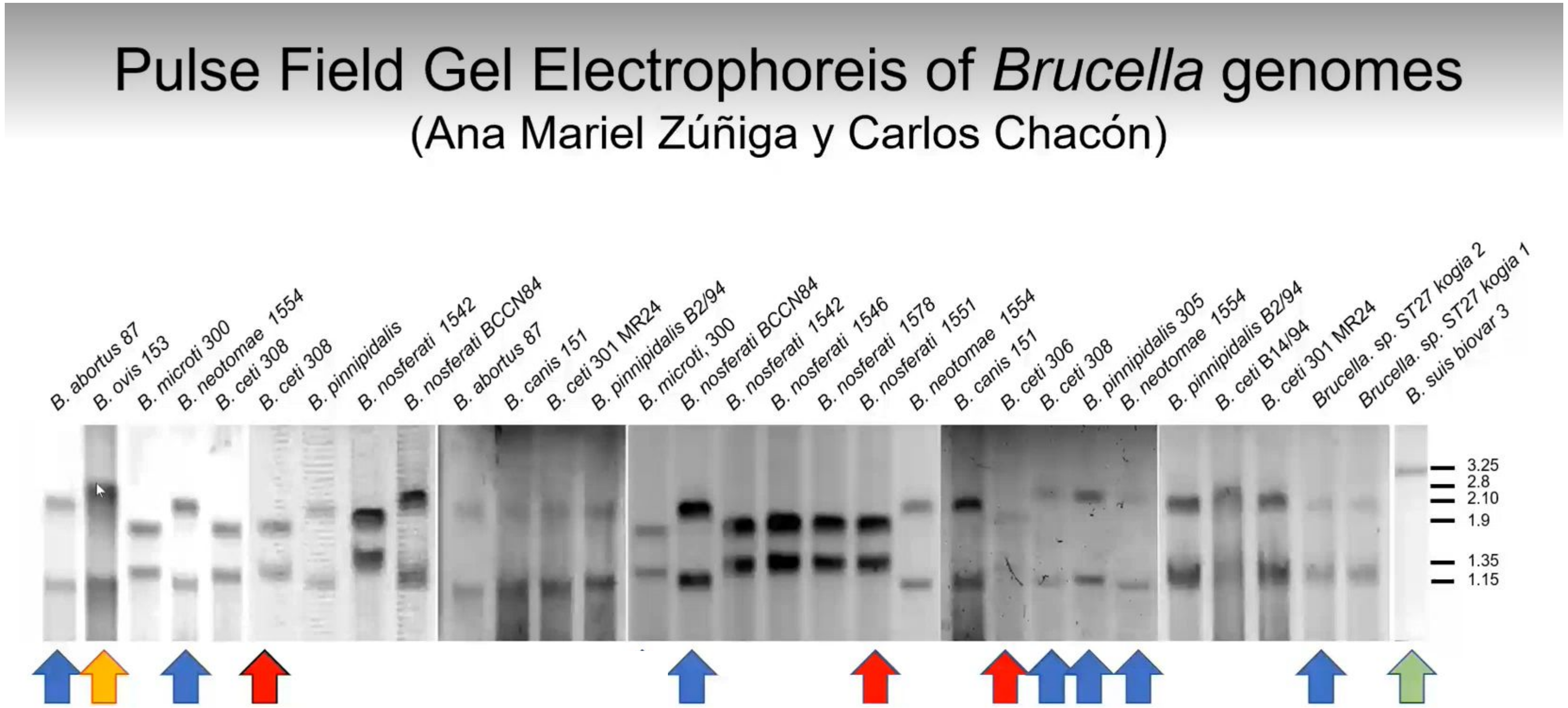
Yuya Kiguchi, Nagisa Hamamoto, Yukie Kashima, Lucky R. Runtuwene, Aya Ishizaka, Yuta Kuze, Tomohiro Enokida, Nobukazu Tanaka, Makoto Tahara, Shun-Ichiro Kageyama, Takao Fujisawa, Riu Yamashita, Akinori Kanai, Josef S. B. Tuda, Taketoshi Mizutani & Yutaka Suzuki

Nature Communications **16**, Article number: 7397 (2025) | Cite this article

Inocle: Insertion sequence encoded; oral origin; circle genomic structure

(368 kb on average, 245 kb on average)

What is it? A plasmid, a megaplasmid, a chromid?



10% of bacterial genomes have chromids: core + accesory genes, = %GC, plasmid replication and partioning systems.
ASSEMBLY ISSUES…

Thanks!

Thank you Matt Beale & Matt Dorman